Multi-AS IS-IS configuration

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Purpose:

The purpose of this lab is to practice the configuration of Integrated Intermediate System to Intermediate System (IS-IS) for both ipv4 and ipv6, a routing protocol useful in an Internet Service Provider (ISP) setting and is one of the main protocols governing the internet.

Background Information:

Intermediate System to Intermediate System (IS-IS) is a classless, dynamic link-state routing protocol used to exchange routing information within an autonomous system. The IS-IS routing protocol was first proposed by the International Organization for Standardization (ISO). IS-IS was part of the ISO/OSI (Open Systems Interconnection) protocol stack. It was first intended to route CLNP (Connectionless Network Protocol); however, subsequent implementations extended routing capabilities to include IP packets. The extension for IP is known as Integrated IS-IS, and the protocols are interchangeably named. IS-IS is a routing protocol commonly found in somewhat large service provider and enterprise-level networks. It gained such popularity from the protocol's internal scalability, convergence time, and multiple fields of configuration. IS-IS works natively over OSI Layer 2 (data link layer). In comparison, other routing protocols (like OSPF) operate natively over IP, meaning they are dependent on an established IP network for functionality. Essentially, IS-IS is agnostic of IP (even though it can route an IP packet) in determining if it operates/statuses. Its PDU (protocol data unit) is unique, and connectionless control messages are ISO 10589.

IS-IS is a link-state routing protocol. It's an Interior Gateway Protocol (IGP) defined by the International Organization for Standardization (ISO) that allows routers (Intermediate Systems, per ISO terminology) to communicate with one another and share link connection states. Thereafter, each router forms its own database of the connectivity —Link-State Database (LSDB)—of the entire network. Each router can then execute the Shortest Path First (SPF)—Dijkstra's Algorithm—to derive the best path for a particular destination. IS-IS routers form neighbor adjacencies via the Hello Protocol Data Unit (PDU); through this process, routers learn about and maintain awareness of directly connected link neighbors. Routers send a PDU Hello signal and await a response; if received, the IS-IS routers have formed an adjacency and can effectively communicate

with one another. Thereafter, IS-IS routers use the Link-State Protocol Data Unit (LSP) to disseminate topology updates across the new link. Distance-vector routing protocols assess a distance between routers; therefore, routers and nodes create their routing tables based upon distance information relayed by neighboring routers. In contrast, IS-IS routers create their routing tables based upon the whole topology.

One key aspect of IS-IS is its hierarchical structure, which consists of areas and has three levels of routing: Level 1, Level 2, and Level 1-2. The levels define how routers communicate with each other and how routing information is advertised to each area. Each router in an area can typically only see the routers in that area, which improves scalability by controlling the size of the LSDB and limiting the area where LSDB updates are propagated. Level 1 routers talk only to other Level 1 routers which are in the same area. The Level 1 routers maintain information about other routers only within the area. Level 1 routers only use intra-area LSPs to use to build its Local LSDB and cannot compute paths to destinations outside of the area. When a Level 1 router wants to reach another area, it forwards its traffic to a Level 1-2 router for the destination. This is important, because a Level 1 router cannot reach a router in another area directly i.e. it cannot automatically communicate with any other area or keep track of destinations outside its area. Level 1 routing is a flat address scheme within the area, so all level 1 routers must share the same area address with their directly connected Level 1 routers. Level 1 routing is best applied in an environment where a simple routing scheme is desired, and the isolation of routing information is required.

Level 2 routers are utilized in two keyways to provide an interconnection function across IS-IS areas, similar to an OSPF Area Border Router (ABR) Level 2 routers maintain a Level 2 Link State Database (LSDB) with routing information regarding other areas and the backbone. A Level 2 router does not need to have Level 1 neighbors, although it commonly has in practice. The Level 2 LSDB only includes inter-area information; a Level 2 router can route traffic from one area to another between two Level 1 networks without knowing the topology of each individual Level 1 area. This distinction creates a scalable routing architecture where changes to the topology inside one of the individual areas does not affect the entire network. In an IS-IS topology, all Level 2 routers must remain connected together in a manner that is contiguous to allow full reachability between areas.

Level 1-2 routers have the ability to do Level 1 and Level 2 routing at the same time. Thus, a Level 1-2 router keeps two LSDBs, one for Level 1 (intra-area) routes and one for Level 2 (inter-area) routes. In an IS-IS routed environment, Level 1-2 routers are critically important as the linking device between areas. For example, when a Level 1 router needs to reach a destination outside of its own area, it forwards all its traffic to an appropriate Level 1-2 router, which can then route the traffic using Level 2 to its destination area. Level 1-2 routers advertise a default route into their Level 1 area, permitting all Level 1 routers to forward their unknown destinations to it without needing to be aware of the inter-area

topology. Segregating domain layering permits changes in topology to be limited to a defined area and minimizes the load from topological changes or link-state updates to a natural limit (bound by areas), which is crucial for the scalable design and capacity of large networks.

In a practical sense, the hierarchical levels can intentionally be combined to create effective IS-IS networks. A typical example is the “ideal” of building core routers as Level 2 only while the edge routers that connect end-user networks are Level 1 only. Routers at the border between the core and edge are configured as Level 1-2 routers. This is like how OSPF builds backbone areas and stub areas to build efficient and low-complexity routing. For example, a large ISP network could define several Level 1 areas to encompass the different geographic or functional areas; routers in these Level 1 areas would only know the topology of neighboring routers and the local topology. Level 1-2 would be routers at the edges of each Level 1 area that then connect to the Level 2 backbone. In such a case, a topology change in one city (Level 1 area) does not cause the entire network to recompute its topology; rather, it only causes the specific Level 1 area to be recomputed, which ensures the network remains stable and scalable.

The area address has a big impact on deciding the routing level of each router when setting up IS-IS. Routers that share an area address will try to create Level 1 connections. Routers without a common area address will try to make Level 2 connections. Level 1-2 routers aim to form both types of connections based on what their neighbors can handle. To set up an IS-IS router, you need to turn on IS-IS on interfaces, set the router's level, and give it a network entity title (NET). The NET includes the area ID and system ID, which help figure out area membership and how connections form.

The key advantages that IS-IS has over other IGPs (Interior Gateway Protocols) such as EIGRP or OSPF come down to 3 crucial differences, first, it was designed as a layer 2 protocol for CLNP (Connectionless Network Protocol) which means that it supports any protocol without any additional configuration – as long as it is above layer 2, this is why it supports ipv4 and ipv6 without any additional configuration, and with this strong flexibility over protocol, it is able to be more adaptable than these other protocols and can be quite helpful to simplify complex configurations, second, its route redistribution in zones, where level 1 routers can only communicate with level 1, 1-2 with both, and 2 with only 2, this in itself isn’t terribly important these are logical levels, as opposed to OSPF in which there must be physical or virtual connection between routers in order for them to communicate, this massively simplifies router placement and configuration as the physical router location is agnostic to network function. Finally, there are less route updates sent compared to EIGRP and OSPF reducing overhead. It is for these reasons that IS-IS is incredibly popular in large scale ISP core networks.

Summary:

In this lab I configure 6 routers each with IS-IS routing protocol enabled to allow communication across the network, Routers 1 and 6 only run level 1, routers 2 and 5 run level 1-2 and routers 3 and 4 run only level 2. This splits the network into 3 autonomous systems divided along the lines of 1-2, 3-4, and 5-6. This represents a typical ISP setup in which, there are different autonomous systems to simplify route calculation

Commands:

hostname r# - Sets hostname
ipv6 unicast-routing – allows ipv6 configuration
router isis CORE – enables IS-IS
net 49.0001.0000.0000.0001.00 - the 49 is the country/region code – the 0001 represents the division such as a campus or college – the next 4 digits are placeholders for departments and the following 4 are for a sub-department – the 0001 represents the unique entity – last 00 is version or modifier.
is-type level-1 – sets the IS-type to level 1, 1-2, 2
metric-style wide – sets the metric style to wide
address-family ipv4 unicast – creates an address-family for ipv4
redistribute connected – redistributes connected routes through IS-IS
redistribute static – used on outside routers to show static connections to router
address-family ipv6 unicast – sets ipv6 address family
redistribute connected – redistributes ISIS routes
redistribute static – redistributes statically connected routes
interface G0/0/0 - enters interface config
description --> R#/Host# (right) - sets description for where the interface is facing
ip address 192.168.0.5 255.255.255.252 - sets ipv4 address
ipv6 address 2001:db8:acad:2::1/64 - sets ipv6 address
ip router isis CORE – enables isis on interfaces
ipv6 router isis CORE – enables ipv6 isis on interface
no shutdown – turns on the port
interface G0/0/1 - enters interface config
description <-- R#/Host# (left) - sets description of where the interface is facing
ip address 192.168.0.1 255.255.255.252 - sets ipv4 address
ipv6 address 2001:db8:acad:1::1/64 - sets ipv6 address
ip router isis CORE – enables ipv4 isis address
ipv6 router isis CORE – enables ipv6 address
no shutdown – turns on the port

Configurations:

Router 1:

```
hostname R1
ipv6 unicast-routing
router isis CORE
 net 49.0001.0000.0000.0001.00
 is-type level-1
 metric-style wide
 address-family ipv4 unicast
  redistribute connected
  redistribute static
 address-family ipv6 unicast
  redistribute connected
  redistribute static
interface Gig0/0/0
 description --> R2 (right)
 ip address 192.168.0.5 255.255.255.252
 ipv6 address 2001:db8:acad:2::1/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
interface Gig0/0/1
 description <-- Host (left)
 ip address 192.168.0.1 255.255.255.252
 ipv6 address 2001:db8:acad:1::1/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
```

ROUTER 2:

```
hostname R2
ipv6 unicast-routing
router isis CORE
 net 49.0001.0000.0000.0002.00
 is-type level-1-2
 metric-style wide
 address-family ipv4 unicast
  redistribute connected
 address-family ipv6 unicast
  redistribute connected
interface Gig0/0/1
 description <-- R1 (left)
 ip address 192.168.0.6 255.255.255.252
 ipv6 address 2001:db8:acad:2::2/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
interface Gig0/0/0
 description --> R3 (right)
 ip address 192.168.0.9 255.255.255.252
 ipv6 address 2001:db8:acad:3::1/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
```

ROUTER 3:

```
hostname R3
ipv6 unicast-routing
router isis CORE
 net 49.0002.0000.0000.0003.00
 is-type level-2
 metric-style wide
 address-family ipv4 unicast
  redistribute connected
 address-family ipv6 unicast
  redistribute connected
interface Gig0/0/1
 description <-- R2 (left)
 ip address 192.168.0.10 255.255.255.252
 ipv6 address 2001:db8:acad:3::2/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
interface Gig0/0/0
 description --> R4 (right)
 ip address 192.168.0.13 255.255.255.252
 ipv6 address 2001:db8:acad:4::1/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
```

ROUTER 4:

```
hostname R4
ipv6 unicast-routing
router isis CORE
 net 49.0002.0000.0000.0004.00
 is-type level-2
 metric-style wide
 address-family ipv4 unicast
  redistribute connected
 address-family ipv6 unicast
  redistribute connected
interface Gig0/0/1
 description <-- R3 (left)
 ip address 192.168.0.14 255.255.255.252
 ipv6 address 2001:db8:acad:4::2/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
interface Gig0/0/0
 description --> R5 (right)
 ip address 192.168.0.17 255.255.255.252
 ipv6 address 2001:db8:acad:5::1/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
```

ROUTER 5:

```
hostname R5
ipv6 unicast-routing
router isis CORE
 net 49.0003.0000.0000.0005.00
 is-type level-1-2
 metric-style wide
 address-family ipv4 unicast
  redistribute connected
 address-family ipv6 unicast
  redistribute connected
interface Gig0/0/1
 description <-- R4 (left)
 ip address 192.168.0.18 255.255.255.252
 ipv6 address 2001:db8:acad:5::2/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
interface Gig0/0/0
 description --> R6 (right)
 ip address 192.168.0.21 255.255.255.252
 ipv6 address 2001:db8:acad:6::1/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
```

ROUTER 6:

```
hostname R6
ipv6 unicast-routing
router isis CORE
 net 49.0003.0000.0000.0006.00
 is-type level-1
 metric-style wide
 address-family ipv4 unicast
  redistribute connected
  redistribute static
 address-family ipv6 unicast
  redistribute connected
  redistribute static
interface Gig0/0/1
 description <-- R5 (left)
 ip address 192.168.0.22 255.255.255.252
 ipv6 address 2001:db8:acad:6::2/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
interface Gig0/0/0
 description --> Host (right)
 ip address 192.168.0.25 255.255.255.252
 ipv6 address 2001:db8:acad:7::1/64
 ip router isis CORE
 ipv6 router isis CORE
 no shutdown
```

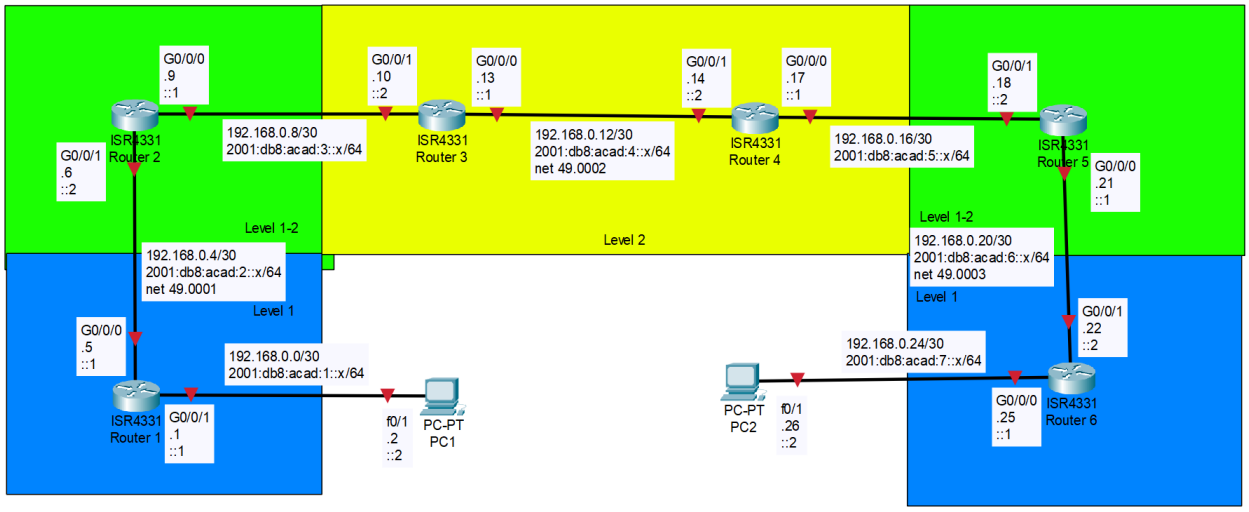
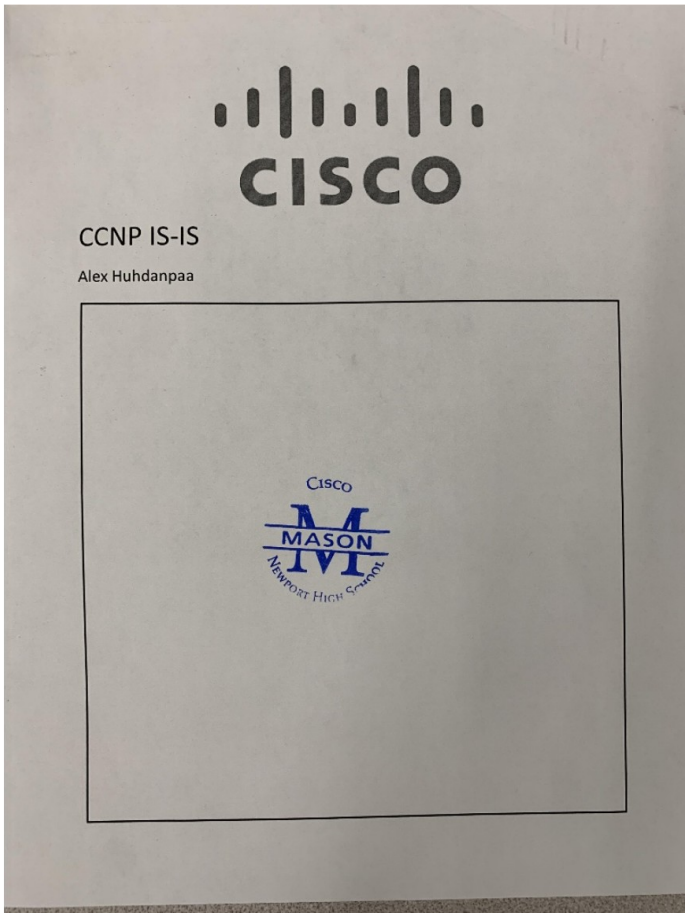
Problems:

The only problem we had was when our configurations for routers 1 and 6 originally had a no in front of the {`ip router isis CORE}` and {`ipv6 router isis CORE}` commands for the outside interfaces, meaning that no information about end-user routes were visible to the LSDB (Link-State Database), however this issue was quickly found in the configuration files and changed

Conclusion:

This lab was a quick and easy success, the configurations were simplified by the lack of need for DHCP (Dynamic Host Configuration Protocol) or internet access, and although we originally made a small slip up on our configurations, we were still able to finish this relatively trivial lab within a day

Diagram and Addressing:



Signoff: